22 (a) Complete this table of values for $y = x^2 + x - 4$

x	–3	–2	–1	0	1	2	3
y	2	–2	–4	–4	–2	2	8

$x = -3$: $9 - 3 - 4 = 2$

$x = 0$: $0 + 0 - 4 = -4$

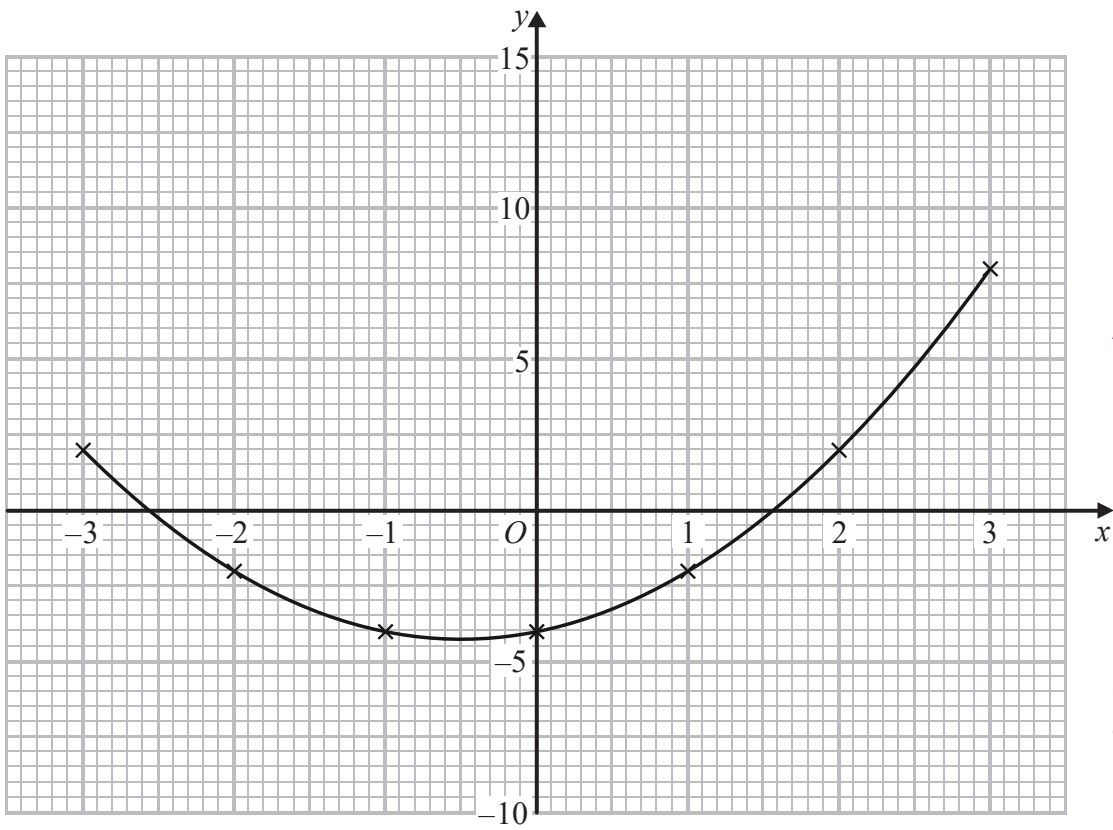
$x = 2$: $4 + 2 - 4 = 2$

$x = 3$: $9 + 3 - 4 = 8$

2 marks total
All four missing values correct

(2)

(b) On the grid, draw the graph of $y = x^2 + x - 4$ for values of x from –3 to 3



1 mark
At least 5 points

1 mark
Smooth curve, below –4

(2)

(c) Use the graph to estimate a solution to $x^2 + x - 4 = 0$

Read where the curve crosses $y = 0$.

1.6

1 mark
One correct graph reading

(1)

(Total for Question 22 is 5 marks)